

Forecasting Stock Returns with Bayesian Econometrics and Machine Learning

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Abstract

This proposal studies the forecasting of excess stock returns through a time-varying parameter (TVP) predictive regression with stochastic volatility (SV) and global–local shrinkage priors, optionally integrated with machine learning (ML) learners. The goal is to improve robustness and adaptivity in environments where static linear models fail due to weak and unstable predictive relations. The framework combines Bayesian state-space estimation with selective regularization via the regularized horseshoe prior, allowing coefficients to evolve slowly and sparsely over time. The model extends to correlated drifts, heavy-tailed observation errors, and limited nonlinearities captured through interaction terms or spline-based single-index components. Empirically, forecasts are evaluated using market and sector excess returns, standard macro-financial predictors, and technical indicators. Out-of-sample performance is assessed through R^2_{OOS} , log predictive score, and CRPS metrics, with forecasts translated into stabilized portfolio allocations. Finally, ML integration is explored through stacking (Elastic Net, Gradient Boosting) and Bayesian residual learning to capture nonlinear signals while preserving interpretability. The overall contribution lies in unifying Bayesian econometrics and machine learning for dynamic, uncertainty-aware stock return prediction and portfolio decision-making.

1 Introduction and Literature Review

We study forecasting of (excess) stock returns using a *time-varying-parameter* (TVP) predictive regression with *stochastic volatility* (SV) and *global–local shrinkage* priors, optionally stacked with light nonlinear ML learners. This line of work responds to weak, unstable predictive relations and time-varying uncertainty in returns. Empirically, static linear models struggle out-of-sample (Welch and Goyal, 2008), while TVP and Bayesian shrinkage improve robustness and adaptivity (Dangl and Halling, 2012; Belmonte et al., 2014; Clark and Ravazzolo, 2015; Pettenuzzo and Timmermann, 2017). Global–local priors such as the horseshoe (Carvalho et al., 2010) or dynamic variants provide aggressive yet selective regularization, guarding against overfitting in high-noise settings. For broad predictive density modeling with many macro predictors, Bayesian VARs/factor models with SV are effective (Koop and Korobilis, 2012). On the ML side, elastic net and gradient boosting often complement Bayesian cores in capturing mild nonlinearities/interactions while remaining relatively stable.

2 Data

We use monthly excess returns for the market and sectors as targets. Predictors include: valuation ratios (Dividend-Price DP, Earnings-Price EP), term and credit spreads, short rate, infla-

tion/breakevens¹, macro growth (industrial production, consumption), as well as technical variables (12–1 momentum, short-term reversal, realized volatility, illiquidity). For cross-sections, we add firm characteristics (size, value, profitability, investment, residual momentum), with hierarchical pooling by GICS sectors if needed. The sample is split into an initial estimation window and a rolling out-of-sample (OOS) evaluation window; at each step t we form h -step-ahead forecasts (typically $h \in \{1, 3, 12\}$ months).

Handling predictor collinearity (implemented). Highly correlated predictors (rates/spreads/BEI/macro) impair identification and inflate uncertainty. We implement two complementary fixes:

- **Groupwise orthogonalization.** Partition predictors into blocks $\mathcal{G}$ (e.g., rates/spreads/inflation). For each block pick an anchor z_t (e.g., 10y yield). For $x_t^{(k)}$ in the block, regress $x_t^{(k)} = a_k + b_k z_t + u_t^{(k)}$ on the expanding window up to t and replace $x_t^{(k)}$ by the residual $u_t^{(k)}$ in the model. This preserves within-block information but removes collinearity with the anchor.
- **Dynamic factors (state-space PCA).** For a block $x_t^{\mathcal{G}} \in \mathbb{R}^{K_G}$, estimate $x_t^{\mathcal{G}} = \Lambda f_t + u_t$ with $f_t \in \mathbb{R}^r$, $r \ll K_G$, and $f_t = A f_{t-1} + \xi_t$. Use f_t as predictors in place of $x_t^{\mathcal{G}}$. Loadings Λ allow interpretation of which variables drive each factor.

3 Empirical Strategy

3.1 Model

Let r_{t+1} denote the excess return and $x_t \in \mathbb{R}^K$ the predictor vector at t (standardized). Our baseline predictive density model is given in Eq. (26). For clarity:

$$r_{t+1} = x_t^\top \beta_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, h_t), \quad (1)$$

$$\beta_t = \rho \odot \beta_{t-1} + \omega_t, \quad \omega_t \sim \mathcal{N}(0, W), \quad (2)$$

$$\log h_t = \mu_h + \phi_h(\log h_{t-1} - \mu_h) + \sigma_h \eta_t, \quad \eta_t \sim \mathcal{N}(0, 1). \quad (3)$$

Why AR(1) on β_t ? Setting $\rho \in (-1, 1]^K$ addresses the confusion between level and drift: RW is the limiting case $\rho \equiv 1$; allowing $\rho < 1$ forces coefficients to revert unless the data support persistent changes. This curbs spurious meandering.

3.1.1 Alternative coefficient dynamics

AR(1) coefficients (baseline model hereafter).

$$\beta_{t,j} = \rho_j \beta_{t-1,j} + \omega_{t,j}, \quad \rho_j \sim \text{Beta prior mapped to } (-1, 1), \quad \omega_{t,j} \sim \mathcal{N}(0, W_{jj}). \quad (4)$$

Concentrate prior mass on $\rho_j \in [0.8, 0.98]$ to allow slow but mean-reverting evolution.

Sparse increments: piecewise constant with jumps (robustness alternative to baseline).

$$\Delta \beta_{t,j} \equiv \beta_{t,j} - \beta_{t-1,j} \sim (1 - \pi_j) \delta_0 + \pi_j \mathcal{N}(0, \widetilde{W}_{jj}), \quad (5)$$

$$\pi_j \sim \text{Beta}(a_\pi, b_\pi), \quad a_\pi \ll b_\pi. \quad (6)$$

Most increments are exactly zero; occasional jumps capture regime changes cleanly. This resolves the *drift vs level* confounding by making change events sparse.

¹Breakeven inflation is a market-implied inflation proxy from nominal vs. real Treasury yields (Treasury Inflation-Protected Securities TIPS): $BEI_t^n = y_t^{nom,(n)} - t_t^{TIPS,(n)}$ for the same maturity n (e.g., 5y, 10y). It embeds risk and liquidity premia.

3.1.2 Heavy-tail specification and calibration

We avoid stacking multiple tail mechanisms. Start with *Gaussian observation errors* and SV (Eqs. (1)–(3)). If PIT tails *under-cover*, switch to *Student-t observation errors* via a scale mixture:

$$\varepsilon_{t+1} \mid \lambda_t \sim \mathcal{N}(0, h_t/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2), \quad (7)$$

$$\Rightarrow \varepsilon_{t+1} \sim t_\nu(0, h_t). \quad (8)$$

Do not simultaneously use heavy-tailed observation errors *and* heavy-tailed SV shocks unless tail calibration still fails. Choose the simplest model that passes coverage tests.

3.1.3 Nonlinearity beyond linear observation equation

A purely linear mean $x_t^\top \beta_t$ forces state dependence to be mimicked by moving coefficients. We add *targeted* nonlinear terms while keeping conditional Gaussianity:

$$r_{t+1} = x_t^\top \beta_t + (x_t \otimes x_t)^\top \gamma + g(z_t) + \varepsilon_{t+1}, \quad (9)$$

where (i) we include a small set of **economically motivated interactions** inside $x_t \otimes x_t$ (e.g., value×volatility-regime), (ii) $z_t = \theta^\top x_t$ is a learned single index and $g(\cdot)$ is a spline basis (B-splines) with shrinkage on γ, θ and on spline coefficients. All new coefficients can follow AR(1) or sparse-increment dynamics with stronger shrinkage than the main effects.

3.2 Filtering interpretation (prediction–update)

To clarify the role of W , write the filtered mean/covariance $(m_{t|t}, P_{t|t})$ of the coefficients. Prediction step: $m_{t|t-1} = m_{t-1|t-1}$, $P_{t|t-1} = P_{t-1|t-1} + W$. Update step uses $v_{t+1} = r_{t+1} - x_t^\top m_{t|t-1}$ and $S_t = x_t^\top P_{t|t-1} x_t + h_t$ with Kalman gain $K_t = P_{t|t-1} x_t / S_t$ and

$$m_{t|t} = m_{t|t-1} + K_t v_{t+1}, \quad P_{t|t} = P_{t|t-1} - K_t x_t^\top P_{t|t-1}. \quad (10)$$

States β_t can be reported in filtered/predicted/smoothed forms.

3.3 Shrinkage priors (global–local)

We control parameter drift via $W = \text{diag}(W_{jj})$ with a *regularized horseshoe* structure (Carvalho et al., 2010; Piironen and Vehtari, 2017):

$$\omega_{t,j} \sim \mathcal{N}(0, W_{jj}), \quad W_{jj} = \tau^2 \lambda_j^2, \quad (11)$$

$$\log \tau \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \quad \lambda_j \sim \mathcal{C}^+(0, 1), \quad 0 < \tau \leq c_{\text{slab}}. \quad (12)$$

We cap the slab scale c_{slab} and use a prior on $\log \tau$ to avoid “all-shrunk”/“all-free” regimes. A practical calibration sets τ_0 via an *effective model size* m_0 :

$$\tau_0 \approx \frac{m_0}{K - m_0} \cdot \frac{\sigma_\omega}{\sqrt{T}}, \quad (13)$$

where σ_ω is a scale for state innovations and T is the effective sample in the first estimation window.

3.4 Correlated drift structure (beyond diagonal W)

Some coefficients should move together (e.g., rate and spread effects). Replace the diagonal W with a *low-rank plus diagonal* structure on increments $\Delta\beta_t$:

$$\Delta\beta_t \equiv \beta_t - \beta_{t-1} = Lf_t + \epsilon_t, \quad f_t \sim \mathcal{N}(0, I_r \sigma_f^2), \quad \epsilon_t \sim \mathcal{N}(0, D), \quad (14)$$

where $L \in \mathbb{R}^{K \times r}$ with $r \ll K$ and $D = \text{diag}(d_1, \dots, d_K)$. This induces $\text{Var}(\Delta\beta_t) = LL^\top + D$. Place shrinkage priors on L (column-wise) and on D to identify a few *drift factors* that co-move groups of coefficients.

3.5 Regime augmentation: real-time, orthogonalized, and centered

Let $s_t \in \mathbb{R}^G$ be regime weights with $s_{t,g} \geq 0$, $\sum_g s_{t,g} = 1$. To avoid endogeneity/leakage, s_t must be *measurable at t* : either latent and jointly estimated, or computed from *lagged* drivers only. A time-varying transition probability (TVTP) example:

$$\Pr(S_t = g \mid S_{t-1} = g', Z_{t-1}) = \frac{\exp(\alpha_{g'g} + \gamma_{g'g}^\top Z_{t-1})}{\sum_{h=1}^G \exp(\alpha_{g'h} + \gamma_{g'h}^\top Z_{t-1})}, \quad (15)$$

where Z_{t-1} are lagged macro/market instruments. Use either the latent S_t (one-hot) or the implied probabilities s_t (soft). To control collinearity, *center* s_t and orthogonalize the interaction block:

$$\tilde{s}_t = s_t - \bar{s}, \quad \bar{s} = \frac{1}{t} \sum_{u=1}^t s_u, \quad (16)$$

$$\widetilde{(x \otimes s)}_t = \text{residuals of } (x_t \otimes s_t) \text{ on } [\mathbf{1}, x_t, s_t] \text{ (Frisch-Waugh)}, \quad (17)$$

then use

$$r_{t+1} = x_t^\top \beta_t + \underbrace{\widetilde{(x \otimes s)}_t^\top}_{\text{regime interaction}} \delta_t + \varepsilon_{t+1}, \quad \delta_t \text{ with stronger shrinkage than } \beta_t. \quad (18)$$

This makes the main effects interpretable as *average* effects and the interactions as *deviations by regime*. Endogeneity arises if s_t is built with revised data or contemporaneous r_{t+1} ; Eq. (15) prevents that by using only Z_{t-1} .

3.6 Estimation

We treat Eqs. (1)–(3), with the chosen dynamics for β_t and optional nonlinear/interaction terms, as a conditionally Gaussian state-space model and estimate by MCMC:

- Sample $\{\beta_t\}_{t=0}^T$ via FFBS conditional on $h_{1:T}$ and the drift structure (diagonal W or factor form in Eq. (14)).
- Update W using global–local scales $(\tau, \lambda_{1:K})$; use regularized horseshoe hyperpriors and the calibration in Eq. (13).
- Sample SV states $\{\log h_t\}$ with standard SV samplers; update $(\mu_h, \phi_h, \sigma_h)$ from their posteriors.
- If Student- t observation errors are used, sample the local scales $\{\lambda_t\}$ as in Eq. (8).
- Predictors are standardized each step so prior scales are comparable over time.

3.7 Forecasting and Decision Rules

The one-step predictive density is

$$p(r_{t+1} | \mathcal{F}_t) = \int \mathcal{N}(r_{t+1}; x_t^\top \beta_t, h_t) p(\beta_t, h_t | \mathcal{F}_t) d\beta_t dh_t, \quad (19)$$

approximated by Monte Carlo draws. We report predictive mean/quantiles and evaluate OOS via R_{OOS}^2 , LPS, and CRPS (App. A.2).

3.7.1 Stabilized portfolio weights

To avoid weight blow-ups, we replace the naive $\hat{w}_t \propto \hat{\mu}_t / \hat{\sigma}_t^2$ by:

Heuristic stabilizer.

$$\tilde{\sigma}_t^2 = \max\{\hat{\sigma}_t^2, \sigma_{\min}^2\}, \quad \tilde{\mu}_t = \kappa \hat{\mu}_t, \quad \kappa \in (0, 1), \quad (20)$$

$$\hat{w}_t \propto \frac{\tilde{\mu}_t}{\tilde{\sigma}_t^2}, \quad \text{then apply caps and scale to a target vol.} \quad (21)$$

Penalized Markowitz (recommended). Let $\hat{\Sigma}_t$ be a shrinkage covariance (Ledoit–Wolf or factor-SV consistent with marginals). Solve

$$\max_w w^\top \tilde{\mu}_t - \frac{\gamma}{2} w^\top \hat{\Sigma}_t w - \lambda_1 \|w\|_1 - \lambda_2 \|w\|_2^2 - \lambda_T \|w - w_{t-1}\|_1 \quad (22)$$

subject to leverage/position caps and (optionally) no-short constraints. Report CER/Sharpe with trading costs.

3.8 ML Stacking

We retain the original ML integration but enforce *nested* evaluation for stacking weights and, when using ML-as-regressors, we include *out-of-fold* predictions and optionally residualize them against x_t to avoid double counting:

$$r_{t+1} = x_t^\top \beta_t + \sum_{\ell \in \{\text{EN, GB}\}} \gamma_\ell \hat{r}_{t+1|t}^{(\ell)} + \varepsilon_{t+1}, \quad \gamma_\ell \sim \mathcal{N}(0, \sigma_\gamma^2), \quad \sigma_\gamma \text{ small.} \quad (23)$$

Empirical strategy workflow

- 1) **Assemble & preprocess data.** Build monthly excess-return targets (market/sectors), construct predictors (DP, EP, term/credit spreads, rates, BEI, macro growth, momentum/reversal, RVOL, illiquidity; add firm characteristics for cross-sections), standardize features, and split into an initial estimation window plus a rolling OOS window with horizons $h \in \{1, 3, 12\}$.
- 2) **Baseline predictive model (TVP–SV).** Specify the time-varying-parameter regression with stochastic volatility

$$r_{t+1} = x_t^\top \beta_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, h_t), \quad (24)$$

$$\beta_t = \underbrace{\rho \odot \beta_{t-1}}_{\text{AR(1) option}} + \underbrace{\omega_t}_{\text{state innovation}}, \quad \omega_t \sim \mathcal{N}(0, W), \quad (25)$$

$$\log h_t = \mu_h + \phi_h(\log h_{t-1} - \mu_h) + \sigma_h \eta_t, \quad \eta_t \sim \mathcal{N}(0, 1), \quad (26)$$

where $\rho \in (-1, 1]^K$ (elementwise) and $\rho \equiv \mathbf{1}$ reproduces the classic random-walk (RW) *special case* $\beta_t = \beta_{t-1} + \omega_t$. See Sec. 3.1.1 for alternatives (sparse increments) and why this resolves drift-vs-level confusion.

Heavy tails: two clear options are provided in Sec. 3.1.2: (i) Gaussian observation errors with SV, or (ii) Student- t *observation* errors with SV via a scale-mixture. Do *not* enable multiple tail mechanisms simultaneously unless tail calibration requires it.

- 3) **Shrinkage for drift (horseshoe).** Set $W = \text{diag}(W_{jj})$ with $W_{jj} = \tau^2 \lambda_j^2$ and $\tau, \lambda_j \sim \mathcal{C}^+$ to globally shrink coefficient drift while letting a few coefficients vary when supported; use weakly informative priors for SV parameters. We use the *regularized* horseshoe and calibrate τ (Sec. 3.3).
- 4) **Optional regime augmentation.** Use regime weights s_t (computed in *real time* with lagged drivers) and gate predictors $x_t \odot \tilde{s}_t$ where $\tilde{s}_t = s_t - \bar{s}$ is centered. Estimate regime-specific slopes $\tilde{\beta}_t$ under the same global-local shrinkage. We orthogonalize the interaction block to avoid collinearity (Sec. 3.5).
- 5) **Estimation (MCMC, conditionally Gaussian).** Run FFBS for $\beta_{0:T}$ given $h_{1:T}, W$; sample SV states and $(\mu_h, \phi_h, \sigma_h)$; update $(\tau, \lambda_{1:K})$ (parameter expansion/slice for mixing). For correlated drifts, sample low-rank factors of Σ_Δ (Sec. 3.4).
- 6) **Generate predictive densities.** At each OOS date t , draw from the posterior predictive $p(r_{t+1} \mid \mathcal{F}_t)$ to obtain mean, variance, and quantiles for each horizon.
- 7) **Evaluate forecasts (statistical).** Compute R_{OOS}^2 , log predictive score (LPS), and CRPS across the OOS window to assess calibration and sharpness of the full predictive distribution.
- 8) **Decision layer (portfolio construction).** Map forecasts to positions with stabilized mean-variance weights (Sec. 3.7.1): start from $\hat{w}_t \propto \tilde{\mu}_t / \tilde{\sigma}_t^2$ with a variance floor, mean shrinkage, and caps; or solve a penalized Markowitz problem with turnover and leverage constraints. Report Sharpe/CER with trading costs.
- 9) **ML integration (possible addition).** Either (a) *stack* Elastic Net/LightGBM with non-negative weights chosen by *nested* rolling LPS, or (b) *include ML forecasts as regressors* in the observation equation with tight zero-centered priors (out-of-fold and orthogonalized) to capture residual nonlinear signal without double counting.
- 10) **Rolling workflow.** For each t : re-standardize with expanding stats $\rightarrow$ re-estimate TVP-SV (carry prior/posterior) $\rightarrow$ produce h -ahead predictive draws $\rightarrow$ update optional stacking/decision rules $\rightarrow$ move the window forward.

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A Appendix

A.1 Scale-mixture augmentation

A *scale-mixture augmentation* rewrites a non-Gaussian distribution as a *mixture of Gaussians with a random local scale*, introduces that latent scale into the model, and then conditions on it so everything becomes Gaussian.

Student- t observation errors.

$$\varepsilon_t \sim t_\nu(0, \sigma^2) \iff \varepsilon_t \mid \lambda_t \sim \mathcal{N}(0, \sigma^2/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2). \quad (27)$$

This yields robust residuals while preserving conjugacy conditionally on $\{\lambda_t\}$.

Horseshoe (global–local) as a Gaussian scale mixture. As in the main text, $\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \tau^2 \lambda_j^2)$ with half-Cauchy priors on scales; auxiliary-variable representations enable Gibbs sampling.

A.2 CRPS and LPS

For a predictive CDF F_t and realization y_{t+1} ,

$$\text{CRPS}(F_t, y_{t+1}) = \int_{-\infty}^{\infty} (F_t(z) - \mathbb{I}\{z \geq y_{t+1}\})^2 dz = \mathbb{E}_{X \sim F_t} |X - y_{t+1}| - \frac{1}{2} \mathbb{E}_{X, X' \sim F_t} |X - X'|. \quad (28)$$

With M draws $\{x_m\}_{m=1}^M$:

$$\widehat{\text{CRPS}} = \frac{1}{M} \sum_{m=1}^M |x_m - y_{t+1}| - \frac{1}{2M^2} \sum_{m=1}^M \sum_{n=1}^M |x_m - x_n|. \quad (29)$$

The one-step log predictive score is $\text{LPS}_t = \log f_t(y_{t+1})$ with $f_t = F'_t$.